

Metabolomics and metagenomics reveal the impact of $\gamma\delta$ T inhibition on gut microbiota and metabolism in periodontitis-promoting OSCC

Wei Wei,^{1,2} Jing Li,¹ Boyu Tang,¹ Ye Deng,³ Yan Li,¹ Qianming Chen¹

AUTHOR AFFILIATIONS See affiliation list on p. 14.

ABSTRACT During the process of periodontitis-promoting oral squamous cell carcinoma (OSCC), the periodontitis microbiota can facilitate OSCC development by activating $\gamma\delta$ T cells. Inhibiting $\gamma\delta$ T cells through immunotherapy has been shown to significantly alleviate various types of cancer. However, the underlying mechanism by which inhibiting $\gamma\delta$ T cells influenced cancer treatment has not been fully elucidated. In this study, a mouse model of OSCC with periodontitis was established, and $\gamma\delta$ T cells were inhibited by antibodies. Gut samples from the mice were collected and analyzed by metabolomics, metagenomics, and 16S rRNA. Integrative analysis of the gut metabolome and microbiome revealed that targeting $\gamma\delta$ T resulted in changes in the levels of metabolites associated with cancer in the gut. Although there was no difference in the α diversity of microbiota, β diversity was significantly different, with a more heterogeneous community structure in the mice receiving targeted $\gamma\delta$ T immunotherapy. Statistical analysis of the gut microbiota at the species level revealed a significant enrichment of *Lactobacillus murinus*, which was significantly correlated with $\gamma\delta$ T abundance. The functional analysis revealed that inhibiting $\gamma\delta$ T could impact the functional gene. A comprehensive analysis revealed that *L. murinus* is especially associated with changes in adenine, which also had connection with the concentration of IL-17 and the abundance of $\gamma\delta$ T.

IMPORTANCE This study revealed the effect of $\gamma\delta$ T cells on gut microbial dysbiosis and identify potential links between gut microbiota and metabolism, providing new insights into the role of $\gamma\delta$ T during the process of periodontitis-induced OSCC, and identifying relevant biomarkers for future research and clinical monitoring protocol development.

KEYWORDS $\gamma\delta$ T, OSCC, metabolomics, metagenomics, microbiota

Oral squamous cell carcinoma (OSCC) is a common malignant tumor, with over 90% of cases occurring in squamous tissue of the head and neck. It is characterized by fast metastasis, high recurrence rates, and drug resistance, with over 350,000 cases and ~170,000 deaths from OSCC annually (1, 2). The main risk factors for OSCC include smoking, heavy alcohol consumption, betel nut chewing, and the human papillomavirus (HPV). Additionally, differences in oral microbiota, dietary factors, immune status, environmental pollutants, occupational exposure, and genetic diseases are related to the development of OSCC (3).

Periodontitis is an oral inflammatory disease caused by bacterial plaque microorganisms. In addition to being an oral disease, chronic periodontitis has been shown to develop into a systemic disease (4). Research has shown that there is a correlation between periodontitis and oral cancer (5–7). Periodontitis is one of the risk factors for oral cancer, and patients with periodontitis have a significantly increased risk of

Editor Jack A. Gilbert, University of California San Diego, San Diego, California, USA

Address correspondence to Yan Li, feifeiliyan@163.com, or Qianming Chen, qmchen@scu.edu.cn.

Wei Wei and Jing Li contributed equally to this article. Author order was determined by drawing straws.

The authors declare no conflict of interest.

See the funding table on p. 14.

Received 31 July 2023

Accepted 18 December 2023

Published 23 January 2024

[This article was published on 23 January 2024 with errors in the Funding table in the PDF. The funding information was corrected in the current version, posted on 31 January 2024.]

Copyright © 2024 Wei et al. This is an open-access article distributed under the terms of the [Creative Commons Attribution 4.0 International license](https://creativecommons.org/licenses/by/4.0/).

developing OSCC (8). Our previous research has also provided strong support for the promotion of OSCC development by periodontitis (9).

$\gamma\delta$ T cells are a unique subset of lymphocytes that have been reported to have anti-tumor or pro-tumor functions in several cancer types and play an important role in immune responses (10). As a key subset in oral barrier immune surveillance, $\gamma\delta$ T cells can recognize various pathogen antigens (11). IL-17⁺ $\gamma\delta$ T cells are the major source of interleukin-17 (IL-17), which has an immunosuppressive effect and can also directly promote cancer progression (12). Our previous research has demonstrated that inhibiting $\gamma\delta$ T cells has a significant therapeutic effect on periodontitis-promoted OSCC, and $\gamma\delta$ T cells can be a key target for treating OSCC promoted by periodontitis (9). Research has shown that gut microbiota can inhibit IL-17 production by cecal $\gamma\delta$ T cells, and metabolites derived from these microbiota can reduce the production of IL-17 by gut $\gamma\delta$ T cells (13). The gut microbiota can shape overall immunity and affect health and disease status, including cancer, at the systemic level (14). The sustained interaction between the gut microbiota and the mucosa regulates inflammation and mediates immune tolerance (15). Changes in the gut microbiota may affect the stability of the oral microbiota composition through the gut-oral axis, and may also affect the occurrence of oral cancer through regulatory effects on the immune system (16).

The mechanism by which inhibiting $\gamma\delta$ T cells promotes treatment of OSCC through periodontitis has not been fully elucidated. In this study, we collected and sequenced gut samples from OSCC mice treated with $\gamma\delta$ T cell immunotherapy (Fig. S1), hoping to uncover the role of $\gamma\delta$ T cells in the gut during the process of periodontitis-promoted OSCC and identify relevant biomarkers for future research and clinical monitoring protocol development.

MATERIALS AND METHODS

Collection and processing of oral saliva samples

This experiment also included four healthy volunteers without periodontitis. All volunteers had no other maxillofacial or serious systemic infectious diseases, had not taken antibiotics, hormones, or antifungal drugs within 3 months, had no history of surgery, radiotherapy, chemotherapy, or pregnancy within the past 6 months. At 9 a.m., all participants were asked to rinse their mouths after fasting for 1 hour, and ~2 mL of non-stimulated saliva was collected. After mixing the saliva samples, they were centrifuged at 8,500 g for 10 minutes, and the lower precipitate was aliquoted and quickly frozen for storage. This study was approved and supervised by the Medical Ethics Committee of West China Hospital of Stomatology, Sichuan University (WCHSIRB-OT 2019-015), and all participants gave informed consent.

Experimental mouse model

Six-week-old male BALB/c mice free of specific pathogens (SPF) were purchased from Chengdu Dossy Experimental Animals Co., Ltd. and were housed in the State Key Laboratory of Oral Diseases at Sichuan University. SPF mice were bred in an environment free of specific pathogens, without common pathogenic infections such as bacteria, viruses, and parasites. The use of SPF mice helps eliminate interference from external pathogens, ensuring more reliable experimental results. The mice were fed with sterile food and water throughout the experiment. All procedures in the animal experiment were performed in accordance with the regulations of the State Key Laboratory of Oral Diseases. This animal study was approved by the Ethics Committee of West China Hospital of Stomatology, Sichuan University (WCHSIRB-D-2019-015).

The AOP-control group (establishment of periodontitis and submucosal injection of squamous cell carcinoma for 21 days, advanced OSCC with periodontitis, $n = 6$) and AOP-treated group (inhibition of $\gamma\delta$ T cells by intraperitoneal injection of UC7-13D5 in the AOP-control group, $n = 5$) mice were subjected to periodontitis model treatment:

mice were given sterile water containing kanamycin (0.5 mg/mL) for three consecutive days, and then switched to sterile water without antibiotics after oral bacteria washing. A 5-0 sterile suture was tied around the second molar in the maxilla of the mouse. Microbiota inoculation was obtained from saliva of patients with periodontitis or healthy individuals. We collected 2 mL of saliva from each designated patient or healthy person. All saliva was mixed, aliquoted, and cryopreserved quickly. The frozen human saliva sample was centrifuged at 8,500 g for 15 minutes, and the resulting precipitate was thoroughly mixed with 3% carboxymethyl cellulose, ensuring that each mouse is inoculated with an equal volume of the mixture (9). The mixture was repeatedly applied to the cheek and palatal surfaces of the mouse molar area. This was done for seven consecutive times, with 1-day intervals between each inoculation.

The AOP-control and AOP-treated group mice were all subjected to OSCC model treatment: 5×10^6 SCC7 squamous cell carcinoma cells were resuspended in 50 μ L DMEM medium and injected into the submucosa of the cheek of the mouse. After inoculation of cancer cells, the number of surviving mice was recorded every day. On the 21st day after cancer cell inoculation, the mice were euthanized, and tumor tissue was collected. The AOP-treated group mice were given intraperitoneal injection of TCR $\gamma\delta$ monoclonal antibody (200 mg/mouse) every 2 days until sample collection.

Liquid chromatography-mass spectrometry metabolomics analysis

Under sterile conditions, fecal samples were collected from each group of mice and stored at -80°C for testing. This test was completed by Guangdong Magigene Biotechnology Co., Ltd. ① After grinding and crushing 25 mg of the sample, 500 μ L of extraction solution (40% methanol/40% acetonitrile/10% water) was added. ② The mixture was sonicated for 5 minutes and allowed to stand at -40°C for 1 hour. ③ Under low-temperature conditions, the supernatant was centrifuged at 12,000 g for 15 minutes and transferred to a small bottle for testing. ④ The target compounds were separated by chromatography using the Vanquish ultra-high-performance liquid chromatography system. ThermoFisher Q Exactive HF-X mass spectrometer was used for mass spectrometry data acquisition. ⑤ The original data were converted using ProteoWizard software and annotated with the mass spectrometry database after processing. ⑥ Differential metabolites were visualized in the form of volcano plots and radar charts. By analyzing the enriched pathways and topology of differential metabolites, the pathways with the highest correlation to metabolite differences were identified and displayed in a rectangular tree diagram. The above results were explained in terms of the AOP-treated group compared to the AOP-control group.

16S rRNA sequencing analysis

① Collected oral and fecal samples from each group of mice under sterile conditions and stored them at -80°C for later use. ② Extracted total DNA according to the instructions of the TIANamp Bacteria DNA kit, and measured the purity and concentration of DNA using a NanoDrop spectrophotometer. Stored the extracted DNA at -80°C for later use. ③ Guangdong Magigene Biotechnology Co., Ltd. utilizes the PE250 strategy on the Illumina NovaSeq 6000 platform, employing primers 515F (GTGCCAGCMGCCGCGG-TAA) and A806R (GGACTACVSGGTATCTAAT) for high-throughput sequencing of the V4–V5 region of the bacterial 16S rRNA gene. ④ Sequence processing was performed after library construction. The sequencing results were annotated and clustered using the UPARSE algorithm (17). ⑤ α -diversity analysis, β -diversity analysis, and differential analysis of some species composition were based on the OTU table.

Metagenomic sequencing analysis

① The sequencing was performed by Guangdong Magigene Biotechnology Co., Ltd. Used the ALFA-SEQ DNA Library Prep Kit to generate sequencing libraries and added index codes. Evaluated the library quality using the Qubit 4.0 Fluorometer (Life

Technologies, Grand Island, NY) and the QSEP400 High-Throughput Nucleic Acid Protein Analysis System (Houze Biology Technology Co., Houze Biology Technology, China). After quality control of the library, the library was sequenced on the Illumina Novaseq 6000 platform, producing paired-end reads with a length of 150 base pairs. ② Species annotation was performed by comparing the sequencing data to the NCBI-NR database, and functional annotation was performed by comparing the sequencing data to the KEGG database.

Co-inertia analysis

To help explain the relationship between the microbiome and metabolomics data sets, we integrated them using a multivariate method called co-inertia analysis (CIA) (18). Principal component analysis (PCA) was applied to each individual data set as input to CIA. The sorting of the two data sets with the highest covariation was determined to identify shared biological trends between the two data sets. CIA was performed using the *ade4* package in R (4.2.1).

Pearson correlation analysis

In order to investigate whether there is a correlation between the microbiota in the gut and oral cavity, this study conducted correlation analysis using the major microbial genera in the gut, oral cavity, AOP-control, and AOP-treated groups. A *P* value of less than 0.05 was considered statistically significant, and Pearson correlation analysis was performed using the *Hmisc* package in R (version 4.2.1).

Statistical analysis

The data are presented as mean $\pm$ SD of independent samples. Analysis of variance and Mann-Whitney *U* tests were used for analyzing parametric or non-parametric data, respectively. Student's *t*-test was used for comparing between two groups. *P* < 0.05 was considered statistically significant. The criteria for metabolomics data analysis used a cutoff of *P* < 0.05 for Student's *t*-test and a variable projection importance greater than one for the first principal component of the orthogonal projections to latent structures discriminant analysis (OPLS-DA) model. Correlation analysis was performed by calculating the Spearman correlation coefficient, and a heat map was drawn for variables with $|r| > 0.8$ and *P* < 0.05.

RESULTS

Inhibition of $\gamma\delta$ T affects gut metabolism in periodontitis promoting OSCC

In our prior research (9), we observed that oral microbiota from periodontitis can facilitate OSCC development by activating $\gamma\delta$ T. In the AOP-treated group, which consisted of OSCC mice treated with periodontitis and UC7-13D5 injection to inhibit $\gamma\delta$ T, the tumor weight was significantly decreased in comparison to their respective controls without $\gamma\delta$ T inhibition. Furthermore, the AOP-treated group exhibited a lower $\gamma\delta$ T cell percentage in tumor tissues and lower IL-17 expression in serum in comparison to their respective controls. These OSCC mouse parameters are presented in Table S1.

To further identify key biomarkers for $\gamma\delta$ T-immunotherapy in periodontitis-associated OSCC, this study conducted non-targeted liquid chromatography-mass spectrometry metabolomics analysis on the feces of mice in the AOP-control and AOP-treated groups, which identified 838 and 300 metabolites in positive and negative ion modes, respectively. After comparing the metabolic compositions of the AOP-treated and AOP-control groups, the differentially expressed metabolites were presented in a volcano plot (Fig. 1A), revealing significant differences in 530 metabolites (positive ion mode) and 147 metabolites (negative ion mode). Both positive and negative ion modes were used in this experiment for metabolite analysis. A radar chart was employed to display a set of metabolites with the most significant changes in content (Fig. 1B).

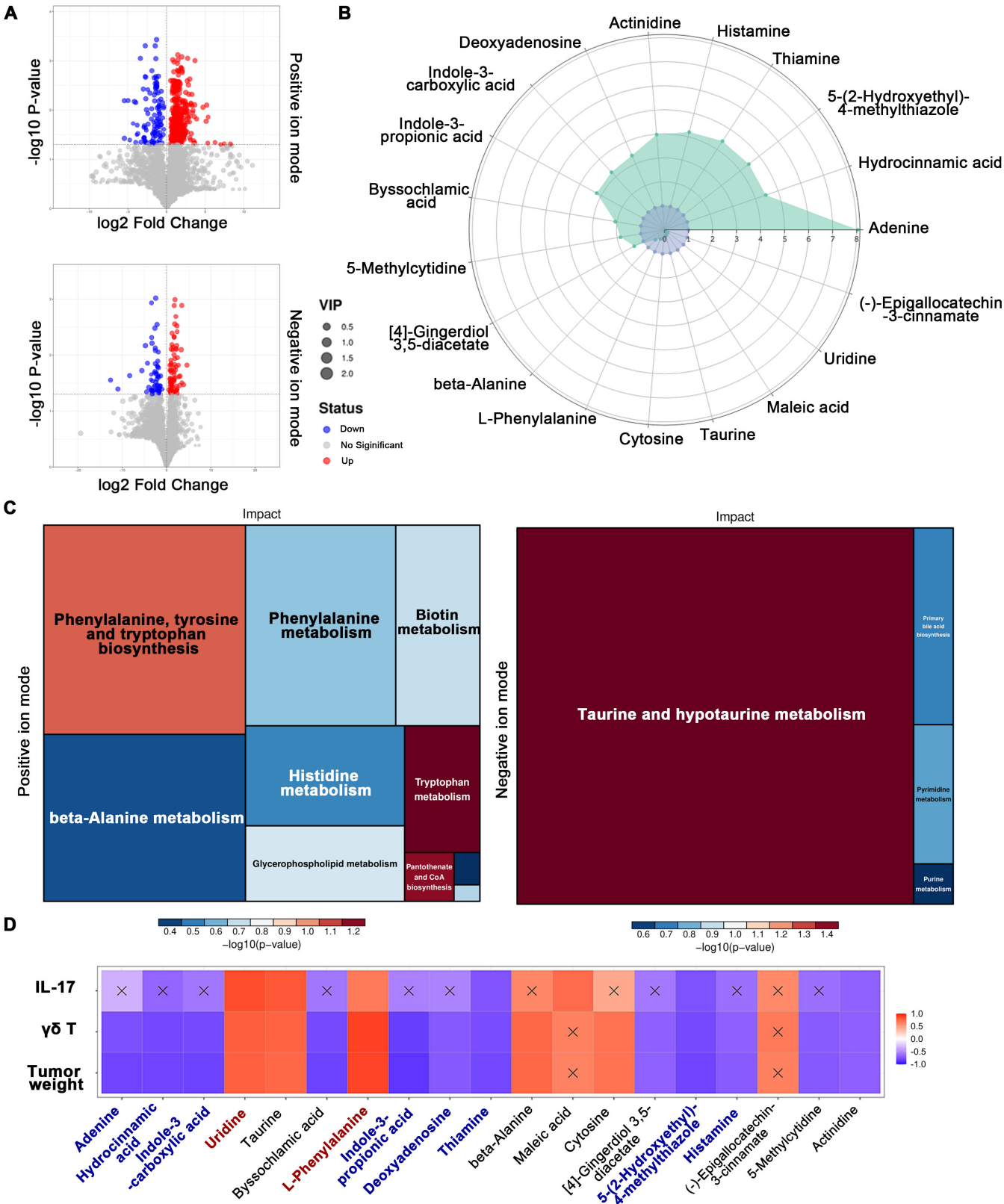


FIG 1 The effects of $\gamma\delta$ T cell inhibition on the fecal metabolome of mice. (A) The volcano plot displays differential metabolites between the AOP-treated and AOP-control groups. The x-axis represents fold changes in each substance, and the y-axis represents the P value of the student t -test. The size of the scatter plot represents the VIP value of the OPLS-DA model. Red dots represent upregulated metabolites, and blue dots represent downregulated metabolites. (B) The (Continued on next page)

FIG 1 (Continued)

radar plot shows trends in changes in the levels of differential metabolites. The ratios of the main differential metabolites were calculated, and a radar plot was made after logarithmic transformation with a base of 2. The purple shadow represents the AOP-control group, and the green shadow represents the AOP-treated group. (C) KEGG pathway enrichment analysis indicates 14 affected metabolic pathways. The size of the square represents the impact factor. The left panel represents the positive ion mode, and the right panel represents the negative ion mode. (D) Correlations between fecal metabolites, IL-17, $\gamma\delta$ T cells, and tumor weight. The correlation coefficient (r value) is represented using a square color map, with red indicating positive correlation and blue indicating negative correlation. The darker the color, the higher the correlation. An "X" in a square indicates that the correlation analysis is not statistically significant.

Among them, the AOP-treated group exhibited a significant increase in the content of adenine, hydrocinnamic acid, 5-(2-hydroxyethyl)-4-methylthiazole, thiamine, histamine, actinidine, deoxyadenosine, indole-3-carboxylic acid, indole-3-propionic acid, byssochlamic acid, 5-methylcytidine, and [4]-gingerdiol-3,5-diacetate, when compared to the AOP-control group. Meanwhile, the content of beta-alanine, L-phenylalanine, cytosine, taurine, maleic acid, uridine, and (-)-epigallocatechin-3-cinnamate decreased significantly. The fecal metabolites adenine (19), hydrocinnamic acid (20), indole-3-carboxylic acid (21), deoxyadenosine (22), thiamine (23), and histamine (24), which are enriched in the AOP-treated mice, have been shown to have inhibitory effects on cell proliferation. Additionally, 5-(2-hydroxyethyl)-4-methylthiazole (25) and indole-3-propionic acid (26) are associated with anti-inflammatory effects. Conversely, the observed reduction of uridine (27, 28) and L-phenylalanine (29) in the feces of AOP-treated mice is associated with an increased risk of cancer.

To gain a better understanding of the metabolic pathways involved in the promotion of OSCC through the oral microbiome in periodontitis by $\gamma\delta$ T cells, significant changes in metabolites were analyzed using KEGG pathway enrichment analysis (Fig. 1C). The results revealed that the inhibition of $\gamma\delta$ T cells in mice with periodontitis-associated OSCC affected a total of 14 metabolic pathways, primarily involving taurine and hypotaurine metabolism, pyrimidine metabolism, primary bile acid biosynthesis, purine metabolism, phenylalanine, tyrosine, and tryptophan biosynthesis, beta-alanine metabolism, phenylalanine metabolism, and biotin metabolism, among others. Of these pathways, cell signaling and environmental stress response were identified as the most relevant biological processes.

This experiment conducted correlation analysis between the identified differential metabolites, the expression of IL-17, the abundance of $\gamma\delta$ T, and tumor weight. The results were visualized using a heatmap (Fig. 1D). The metabolites adenine, hydrocinnamic acid, indole-3-carboxylic acid, deoxyadenosine, thiamine, histamine, 5-(2-hydroxyethyl)-4-methylthiazole, and indole-3-propionic acid, which have been reported to have inhibitory effects on cell proliferation or inflammation, were found to be significantly and negatively correlated with the abundance of $\gamma\delta$ T. Conversely, uridine and L-phenylalanine, which have been associated with a high risk of cancer, showed significant and positive correlation with the abundance of $\gamma\delta$ T. After obtaining the matching information of differentially expressed metabolites in each group, we performed pathway search and regulatory interaction network analysis on the KEGG database of the corresponding species. The results of the regulatory analysis are presented as a network plot (Fig. S2).

In the context of OSCC with periodontitis, targeting $\gamma\delta$ T resulted in a substantial elevation in the levels of metabolites associated with anti-cancer effects and a concomitant decrease in those promoting cancer in the gut. This observation highlights the potential involvement of host metabolism in the immunotherapeutic efficacy of $\gamma\delta$ T cell-targeting strategies and underscores the impact on multiple tumor-related metabolites. These findings further suggested that these metabolites might serve as crucial biological markers for the precise and effective immunotherapy of OSCC in the presence of periodontitis.

Targeting $\gamma\delta$ T improves gut microbiota homeostasis in periodontitis promoting OSCC

In this study, fecal samples from both the AOP-control and AOP-treated groups of mice were subjected to 16S rRNA sequencing, resulting in a data set of 734,467 high-quality sequences and identifying 5,352 OTUs. Although analysis using the Chao1, Observed richness, and Simpson indices revealed no significant differences in α -diversity of gut microbiota between the two groups (Fig. 2A), the PCoA analysis showed that

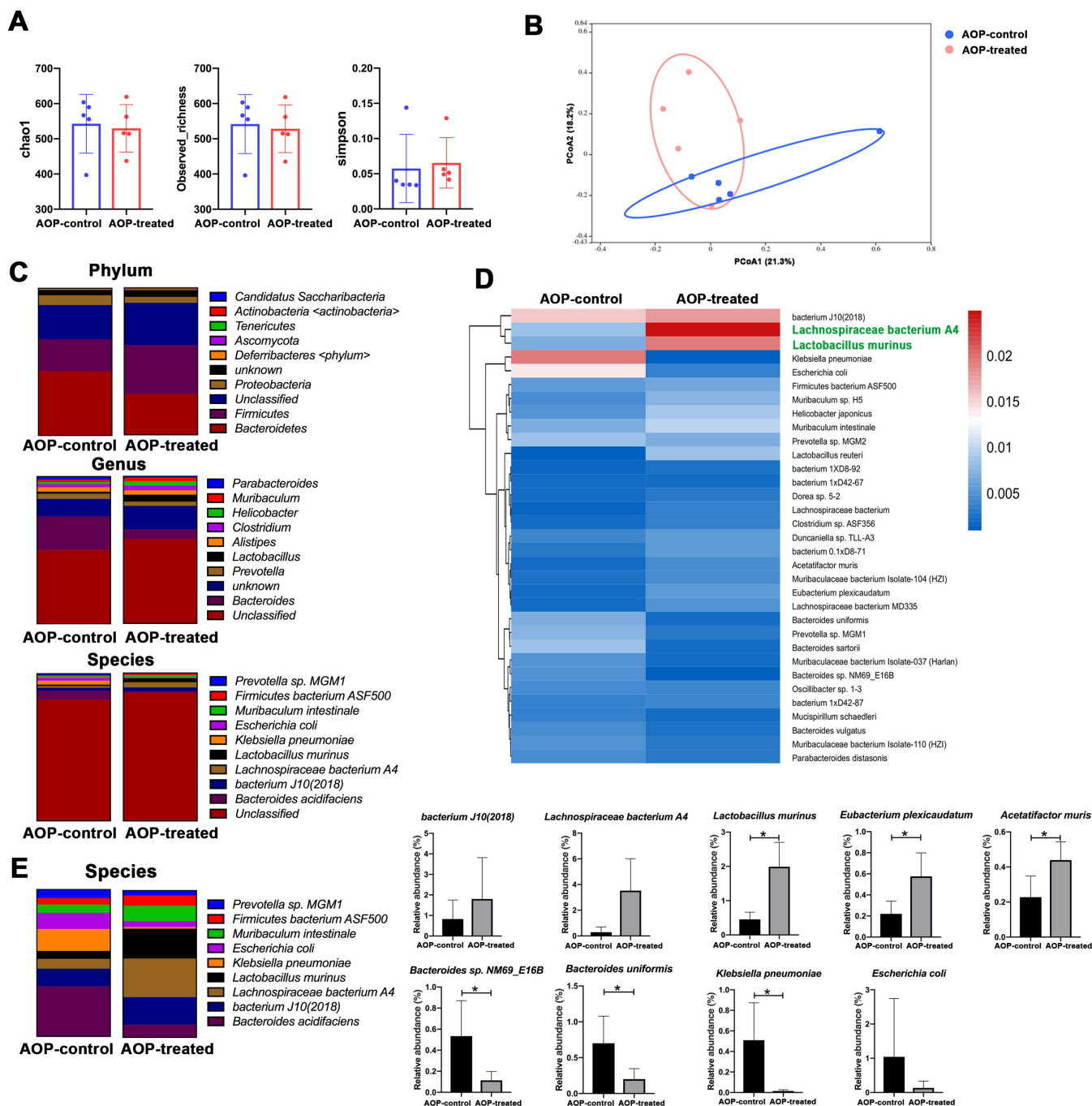


FIG 2 Analysis of diversity and abundance of gut microbiota in mice. (A) α -diversity of gut microbiota in each group of mice, as measured by Chao index, Observed richness, and Simpson index. (B) β -diversity of gut microbiota in each group of mice shown as a PCoA plot of UniFrac distances. (C) Abundance of major phylum, genus, and species of gut microbiota in mice. (D) Heatmap showing the relative abundance of 33 major classified bacterial species within each group. (E) Analysis of inter-group differences in major gut microbiota at the species level. *P < 0.05.

the targeted $\gamma\delta$ T cell immunotherapy had a notable impact on the composition of gut microbiota, with significant differences in β -diversity of gut microbiota observed between the AOP-control and AOP-treated group mice (Fig. 2B).

Furthermore, metagenomic sequencing was conducted to gain further insight into the gut microbiota of the two groups of mice, revealing that the dominant phyla were *Bacteroidetes* (35.4%), *Firmicutes* (27.2%), and *Proteobacteria* (5.2%). At the genus level, *Bacteroides* (12.3%), *Prevotella* (2.6%), *Lactobacillus* (2.5%), *Alistipes* (2.5%), and *Clostridium* (2.3%) were the most abundant genera. Among the species, *Bacteroides acidifaciens* (2.6%), *bacterium J10(2018)* (1.7%), *Lachnospiraceae bacterium A4* (1.7%), *Lactobacillus murinus* (1.3%), *Klebsiella pneumoniae* (1.0%), *Escherichia coli* (0.9%), *Muribaculum intestinale* (0.86%), *Firmicutes bacterium ASF500* (0.6%), and *Prevotella* sp. MGM1 (0.55%) were the most prevalent (Fig. 2C).

Based on the genus-level sequencing results of intestinal microorganisms, the experiment conducted a correlation analysis between all genus-level classifications and the IL-17 protein concentration, $\gamma\delta$ T cell abundance, and tumor weight (Fig. S3). The analysis indicated a significant positive correlation between *Bacteroides* and the abundance of $\gamma\delta$ T, while *Lactobacillus* was significantly negatively correlated with the abundance of $\gamma\delta$ T. During the targeted inhibition of $\gamma\delta$ T in the treatment of OSCC with periodontitis, the abundance of the beneficial intestinal bacteria (30) *Lactobacillus* increased, which was significantly correlated with the IL-17 protein concentration and $\gamma\delta$ T cell abundance.

The study also presented a heatmap that displays the relative abundance of 33 major bacterial species identified in the AOP-control and AOP-treated group. The top three major bacterial species in the gut of mice in the AOP-control group were *bacterium J10(2018)*, *E. coli*, and *K. pneumoniae*, while *bacterium J10(2018)*, *L. bacterium A4*, and *L. murinus* were enriched in the gut of mice in the AOP-treated group (Fig. 2D). Statistical analysis of the gut microbiota at the species level revealed significant differences between the two groups, with the mice receiving targeted $\gamma\delta$ T cell immunotherapy exhibiting a significant enrichment of *L. murinus*, *Eubacterium plexicaudatum*, and *Acetatifactor muris* in the gut, along with a significant decrease in *K. pneumoniae*, *Bacteroides uniformis*, and *Bacteroides* sp. NM69_E16B (Fig. 2E).

Overall, the results suggested that in the promotion of OSCC progression by periodontitis oral microbiota, targeting the inhibition of $\gamma\delta$ T could simultaneously improve gut homeostasis and enrich probiotics.

The impact of inhibiting $\gamma\delta$ T on microbiota functional genes in promoting OSCC by periodontitis oral microbiota

In the context of periodontitis oral microbiota promoting OSCC, inhibiting $\gamma\delta$ T resulted in microbiota functional gene profiles that were almost identical between the two groups, with some differences still present, as shown by the Venn diagram (Fig. 3A). To investigate the specific impact of inhibiting $\gamma\delta$ T on functional genes, KEGG pathway enrichment analysis was performed, which indicated that the pathway distribution of the AOP-control and AOP-treated groups was similar, with metabolic pathways being the largest proportion at the L1 level. Visualization of pathway abundance at the L2 level revealed that both groups had similar abundance levels, with amino acid metabolism, translation, and carbohydrate metabolism exhibiting higher abundance levels (Fig. 3B).

The analysis of the pathways that differed significantly between the AOP-control and AOP-treated groups were performed. The results showed that the AOP-treated group had significantly higher base excision repair, nucleotide excision repair, RNA polymerase, and *Staphylococcus aureus* infection pathways, while the AOP-control group had significantly higher other glycan degradation, lysosome, glycosaminoglycan degradation, pertussis, ascorbate and aldarate metabolism, glycosphingolipid biosynthesis-ganglio series, various types of N-glycan biosynthesis, and monobactam biosynthesis pathways (Fig. 3C). Moreover, this study conducted further analysis of the pathways and key proteins (enzymes) that showed significant differences between the AOP-control and

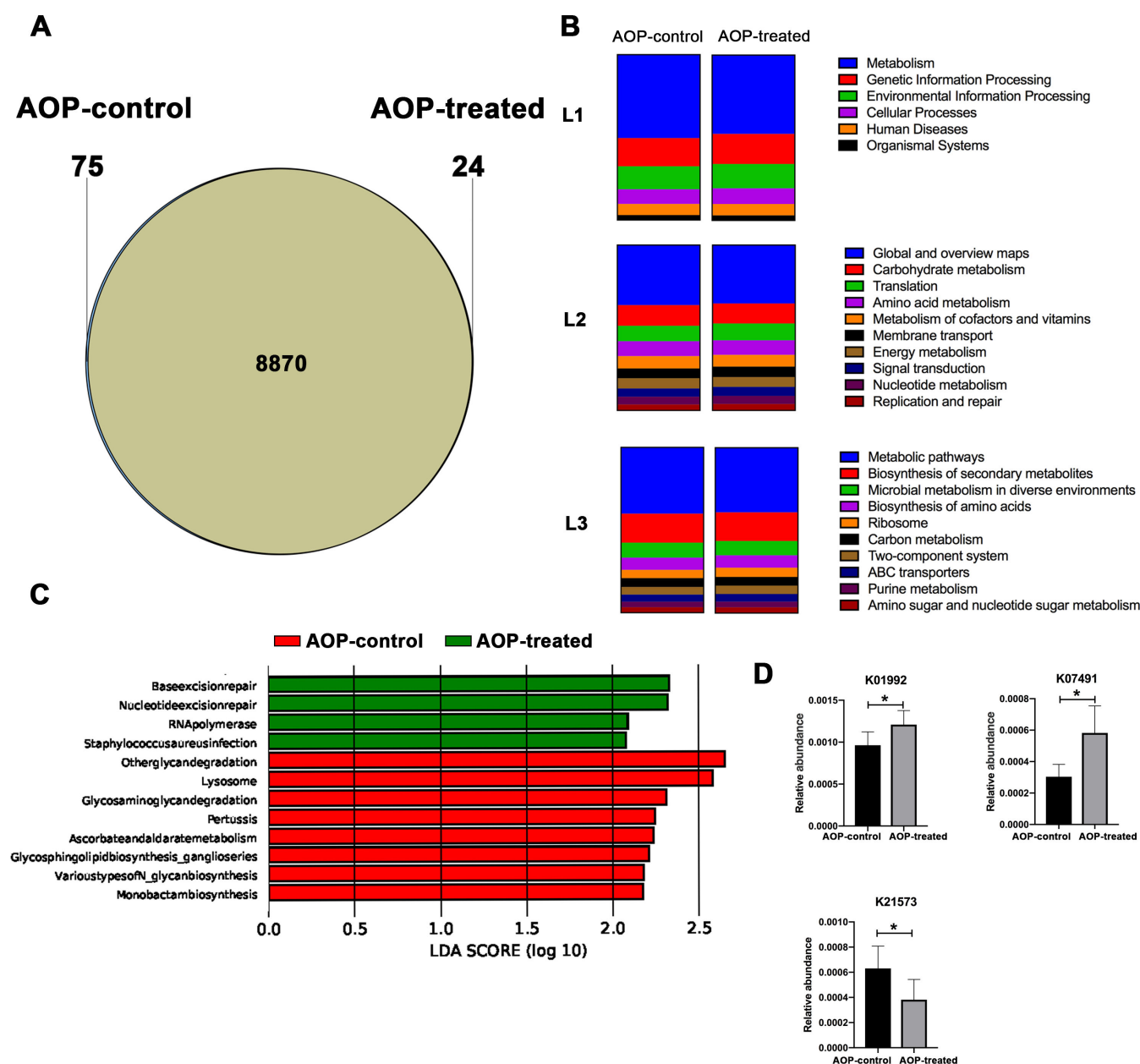


FIG 3 Pathway enrichment analysis and abundance ratio. (A) The Venn diagram intuitively shows the number of common genes among groups. (B) The pathway distribution of AOP-control group and AOP-treated group at different classification levels. (C) The differences between groups in terms of pathways. The graph shows the LDA values of the different species, with colors representing the corresponding groups. The length of the bars in the graph represents the contribution of each species to the differences (LDA Score). LDA score is greater than 2. (D) Analysis of inter-group differences at the KO level. * $P < 0.05$.

AOP-treated groups, revealing differences in the relative abundance of proteins K01992, K07491 (a protein that plays an important role in genome evolution, adaptive evolution, and the immune system) (31, 32), and K21573 between the two groups (Fig. 3D).

Additionally, this study investigated the correlation between these microbiota functional genes and the concentration of IL-17, the abundance of $\gamma\delta$ T, and tumor weight. There were significant negative correlations between the abundance of $\gamma\delta$ T and K07491 (Fig. S4). Moreover, K07491 and K07483 (a transposase that can cause changes in genome structure and gene regulation) (33) were significantly negative correlated with the concentration of IL-17. Furthermore, significant positive correlations were observed between $\gamma\delta$ T and the abundance of K21573 (involved in the degradation and utilization

of starch) (34, 35), K07165 (a type of receptor protein in bacteria that can sense signals in the environment), and K07485 (a type of transposase that can catalyze transposition reactions).

The results indicated that the inhibition of $\gamma\delta$ T cells could impact the functional gene profiles of oral microbiota associated with periodontitis. Furthermore, the observed differences in functional gene profiles and their relationship with the immune system highlight the potential importance of these factors as therapeutic targets for the prevention and treatment of OSCC.

Correlation between gut metabolites and microbiota in periodontitis oral microbiota-promoted OSCC mice with $\gamma\delta$ T suppression

To investigate the potential correlation between the impact of $\gamma\delta$ T suppression on alterations in microbiota and metabolites in the mouse intestine, this study performed bioinformatics analyses on the metabolomic and metagenomic data. PCA was conducted on both data sets, retaining two characteristic values, to visualize the correlation between the two. CIA was then carried out to explore the differences in metabolites and microbiota between the control and experimental groups. This analysis revealed a certain correlation between changes in intestinal metabolism and microbiota at the genus and species levels in both the AOP-control and the AOP-treated mice (Fig. 4A).

The study further investigated the relationship between the main microbiota at the genus or species level and the major metabolites that underwent changes (Fig. 4B and C). A comprehensive analysis revealed that *L. murinus* of the *Lactobacillus* genus, a type of intestinal microbiota, is associated with changes in multiple metabolites, as well as with the concentration of IL-17 protein and the abundance of $\gamma\delta$ T. The key metabolite associated with *L. murinus* is adenine, which has been reported to have potential anticancer effects and is also correlated with the abundance of $\gamma\delta$ T (36).

By conducting bioinformatics analysis, we found that there is a certain correlation between changes in intestinal metabolism and microbiota. Moreover, we identified *L. murinus* as the key microbiota associated with changes in multiple metabolites, the concentration of IL-17 protein, and the abundance of $\gamma\delta$ T. Adenine, a key metabolite associated with *L. murinus*, was also found to have potential as a therapeutic target for preventing and treating OSCC. These findings suggest that targeting *L. murinus* and adenine may offer promising approaches for the treatment of OSCC.

Investigating the impact of inhibiting $\gamma\delta$ T on resistance genes, virulence factors, and microbiota interaction

In this study, we conducted a thorough investigation of the impact of inhibiting $\gamma\delta$ T on resistance genes in periodontitis-associated OSCC. *t*-Tests were performed on the top four highest abundance resistance genes in the AOP-control and AOP-treated groups, and no significant difference in resistance genes was observed between the two groups (Fig. 5A). However, further *t*-test analysis on virulence factors revealed a significant difference in *AslA*, an acidic lipopeptide found in some pathogens (Fig. 5B). *AslA* can degrade important molecules of the host immune system, and help the pathogen evade immune surveillance. Moreover, *AslA* can also cleave important proteins on the surface of host cells, such as platelet-agglutinating factor, interfering with normal host cell function (37, 38).

In addition, we investigated the oral and gut microbiota and found significant differences at the genus level, with *Bacteroides* being the only overlapping genus (Fig. 5C). We also examined the correlation at the genus level between the oral and gut microbiota. In the gut (Fig. 5D), *Bacteroides* was positively correlated with *Parabacteroides* and negatively correlated with *Clostridium*. In the oral cavity (Fig. 5E), the three pairs of *Streptococcus* and *Corynebacterium*, *Staphylococcus* and *Corynebacterium*, and *Escherichia-Shigella* and *Enterococcus* are positively correlated, while *Porphyromonas* and *Citrobacter* are negatively correlated. Furthermore, analysis of the correlation at the genus level of microbiota in the oral cavity of mice with periodontitis-associated

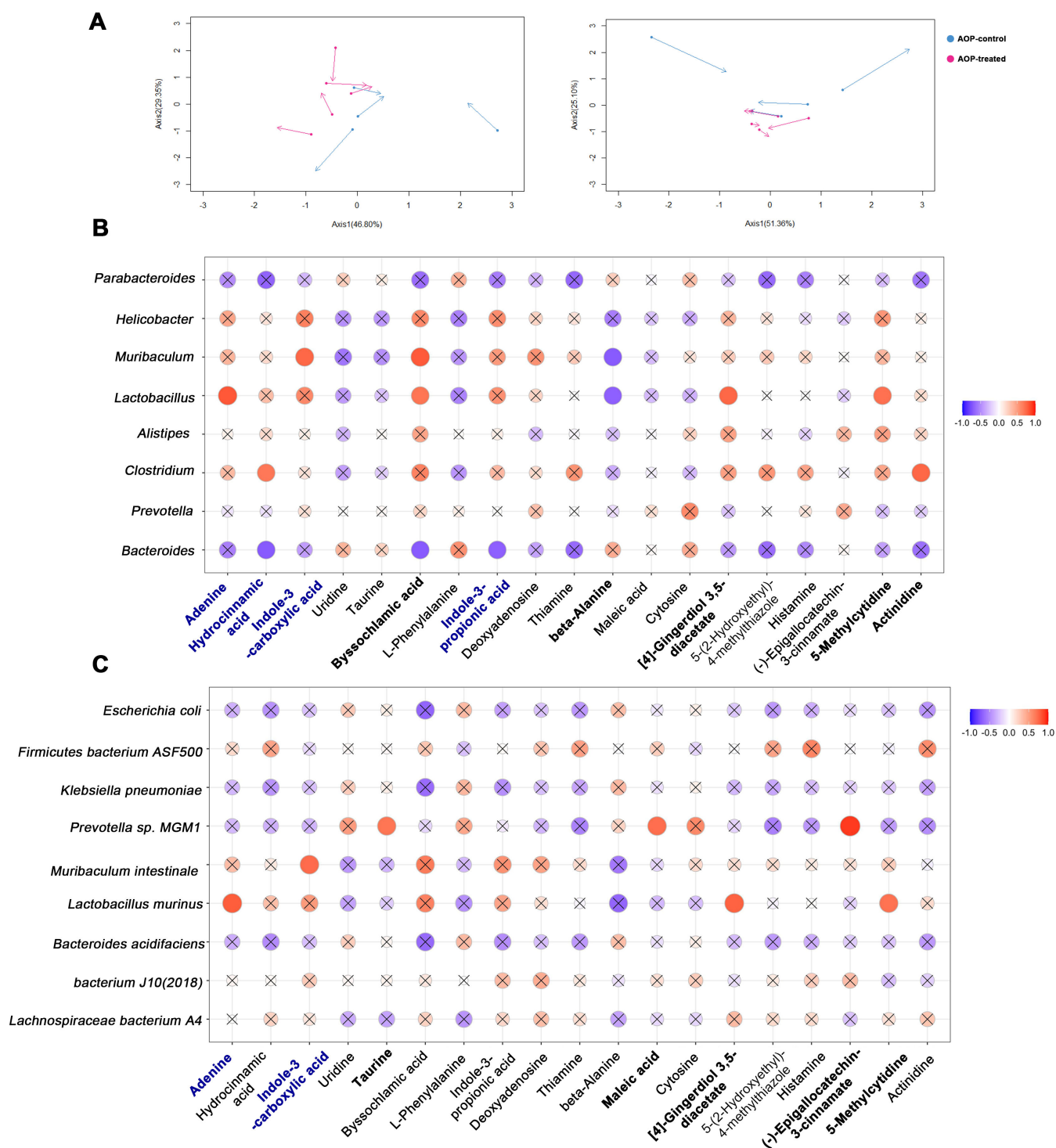


FIG 4 Association between gut metabolites and microbes in mice. (A) The correlation between major microbiota at the genus (left) or species (right) level and major changed metabolites in the intestine is presented, where the starting points represent metagenomic features and the endpoints represent metabolomic features. The blue dots represent the AOP-control group, while the red dots represent the AOP-treated group. (B–C) The correlation between major microbiota at the genus (B) or species (C) level and major changed metabolites in the intestine is shown. The horizontal axis displays the major microbiota, while the vertical axis shows the different metabolites. The strength of the correlation is represented by the color of the circle, with red indicating a positive correlation and blue indicating a negative correlation. The darker the color, the stronger the correlation. An "X" inside the circle indicates that the correlation analysis is not statistically significant.

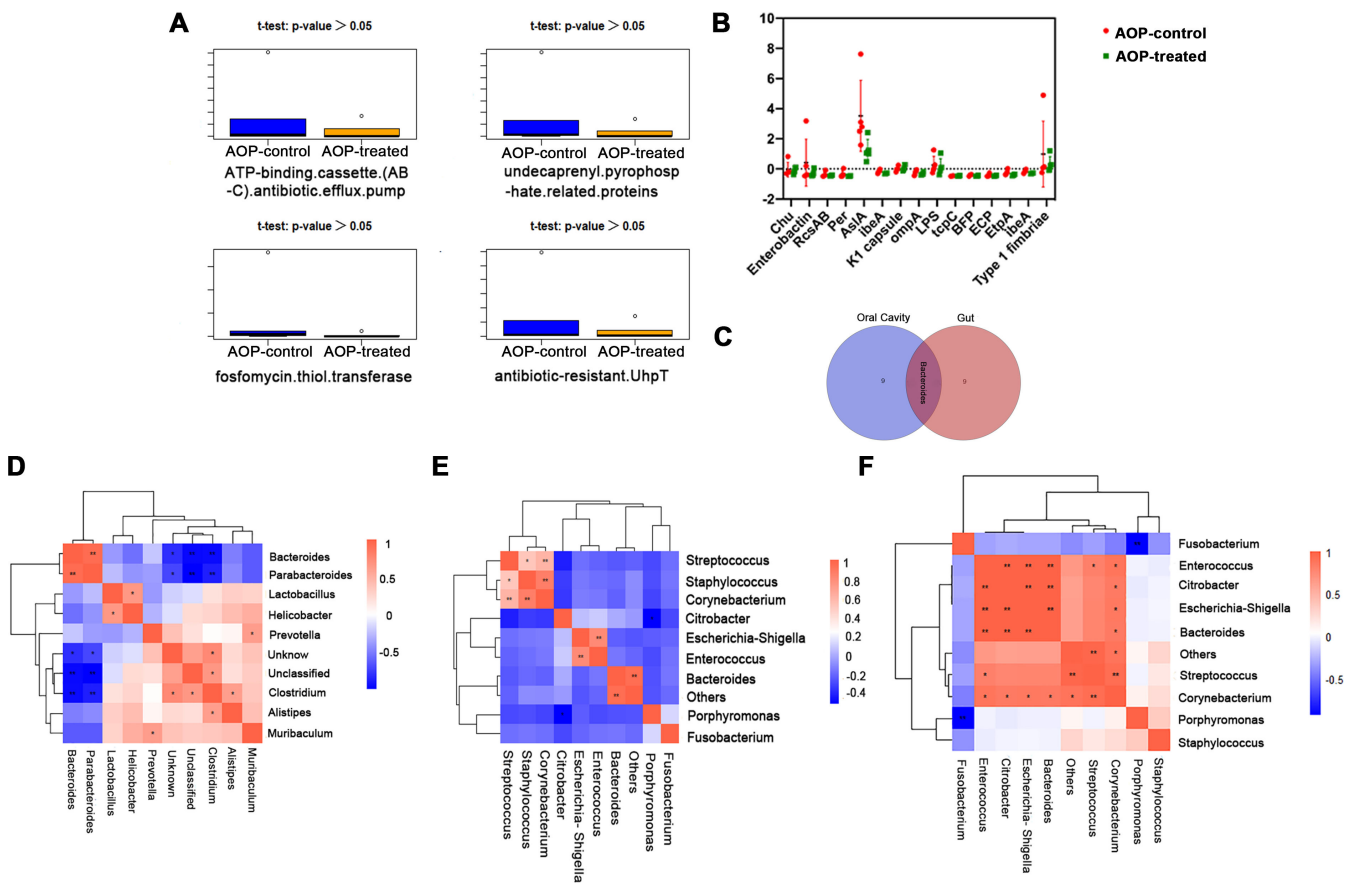


FIG 5 Differences in *t*-tests for mouse resistance genes and virulence factors, as well as at the microbial genus level. (A) *t*-Test of resistance genes between AOP-control group (blue) and AOP-treated group (yellow). (B) *t*-Test of independent bacterial factors. (C) The Venn diagram shows the number of major microbial genera at the level of the mouse gut and oral cavity. (D–F) Correlation analysis of microbial genera at the level of the gut (D) or oral cavity (E) or oral bacteria in the AOP-control group versus the AOP-treated group (F).

OSCC (Fig. 5F) showed that *Bacteroides* was significantly positively correlated with *Enterococcus*, *Citrobacter*, and *Escherichia-Shigella*, *Enterococcus* was positively correlated with *Citrobacter*, *Escherichia-Shigella*, and *Bacteroides*, and *Streptococcus* was positively correlated with *Corynebacterium*.

Our study highlights the significance of inhibiting $\gamma\delta$ T in reducing virulence factors in periodontitis-associated OSCC. Furthermore, our analysis revealed significant differences in the microbiota in the oral cavity and intestine. These findings provide valuable insights into the complex interactions among microbiota and virulence factors, and might contribute to the development of effective treatment strategies for OSCC.

DISCUSSION

$\gamma\delta$ T cells have strong plasticity and can polarize from one phenotype to another. It has been reported that $\gamma\delta$ T cells can polarize into $\gamma\delta$ T17 cells (producing only IL-17), $\gamma\delta$ T1/17 cells (producing both IFN- γ and IL-17), $\gamma\delta$ T1 cells (producing both IFN- γ and TNF- α), and $\gamma\delta$ T2 cells (producing IL-4). $\gamma\delta$ T cells and their secretion of IL-17 play an important role in cancer cell growth (39), while $\gamma\delta$ T cells can also weaken the anti-tumor ability of other immune cells (40, 41). Due to the plasticity of $\gamma\delta$ T cells, they can be regarded as “friend” or “foe” in different cancer contexts.

In certain cancers, $\gamma\delta$ T cells can shape an immune-suppressive tumor microenvironment through their own accumulation and function, or by enhancing the activity of

other immune-suppressive cells, thereby promoting cancer progression. Our previous research has also demonstrated that $\gamma\delta$ T cells are a key upstream node in the pro-carcinogenic effect of periodontal disease-associated oral microbiota.

Previous studies have shown that there is a correlation between the gut microbiota and $\gamma\delta$ T cells. The gut microbiota can inhibit IL-17 production in cecal $\gamma\delta$ T cells, and metabolites derived from these microbiota can reduce IL-17 production by gut $\gamma\delta$ T cells (13). In this study, there was a significant positive correlation between the abundance of $\gamma\delta$ T cells and K07491, and a significant negative correlation with K21573. K07491 is the REP-associated tyrosine transposase (*rayT*), which belongs to the HUH superfamily of nucleases, whose members promote replication or mobilization of various mobile genetic elements (viruses, plasmids, and insertion sequences) (42). In addition to the conserved residues involved in DNA cleavage catalysis, *RAYT* also carries characteristic structural motifs absent from typical IS200/IS605 transposases. In one-third of the cases, the DNA sequence flanking the *rayT* gene is arranged within bacterial interspersed mosaic elements (BIMEs) (32), and recent studies have shown that DNA gyrase and DNA polymerase I can specifically recognize BIME DNA *in vitro*. These results suggest that BIMEs may play a role in the functional organization of bacterial nuclei (43). K21573 is a TonB-dependent starch-binding outer membrane protein *SusC*, which is essential for bacterial growth on starch (35). The process of nutrient acquisition by bacteria is executed by protein machines on the cell membrane. For many bacteria in the gut microbiota, this process is accomplished by a protein complex consisting of a substrate-binding protein (*SusD*) and a channel transporter protein (*SusC*) (34). Therefore, we believe that targeting $\gamma\delta$ T cells as a therapeutic approach can induce changes in the metabolic microenvironment within the intestines.

It is worth noting that $\gamma\delta$ T cells can respond to cytokine stimulation and change their function. Artificially altering the cytokine balance in the tumor immune microenvironment may be another new immunotherapy method targeting $\gamma\delta$ T cells. Researchers advocate routine testing of IL-17 levels before starting $\gamma\delta$ T cell adoptive immunotherapy to predict the immune status. In summary, whether using $\gamma\delta$ T cell expansion or targeting $\gamma\delta$ T cells, attention should be paid to their function and the regulation of cytokine balance in the tumor microenvironment.

Researchers generally agree that there is a mutual interaction between the microbiome and $\gamma\delta$ T cells, which can affect tissue homeostasis and disease development. The role of the microbiome in the development and function of $\gamma\delta$ T cells has been a focus of research. However, the reverse regulation of the microbiome by $\gamma\delta$ T cells has been underestimated by researchers. Although there are reports that lung $\gamma\delta$ T cells in cancer patients are closely positively correlated with lung microbiome alpha diversity (44), the impact of $\gamma\delta$ T cells on the microbiome, especially in the oral microbiome, has not been fully determined. Studies have shown that ablating $\gamma\delta$ T cells significantly alters the diversity of the oral microbiome, and similar microbiome changes have been reported in IL17R-deficient mice (45), supporting the importance of IL-17 signaling in $\gamma\delta$ T cell homeostasis. However, the bidirectional relationship between $\gamma\delta$ T cells and the microbiome, especially the effect of $\gamma\delta$ T cells on the microbiome, has not been fully determined. This study first analyzed the impact of ablating $\gamma\delta$ T cells on gut ecology in promoting OSCC progression by the oral microbiome in periodontitis. The sequencing data of mouse feces supported that $\gamma\delta$ T cells would reshape the host microbiome ecology in a reverse manner. After inhibiting $\gamma\delta$ T cells, the content of anticancer metabolites in the intestines of tumor-bearing mice increased, the content of cancer-promoting metabolites decreased, and the probiotics were enriched in the intestines. Furthermore, it was found that *L. murinus*, adenine, and $\gamma\delta$ T cell abundance were closely related.

In summary, the influence of host microbiota should be considered in the process of targeting $\gamma\delta$ T cell therapy for OSCC with concurrent periodontitis. First, the impact of environmental changes on the reshaping of $\gamma\delta$ T cell function should be carefully considered, including the influence of oral microbiota structure on $\gamma\delta$ T cell numbers and

function, which in turn affects tumor development. In addition, the reverse effect of $\gamma\delta$ T cells on the intestinal ecosystem is also worth exploring and utilizing.

ACKNOWLEDGMENTS

This study was supported by the National Natural Science Foundation of China (81991500, 81991502, 81771085), Key Projects of Sichuan Provincial Department of Science and Technology (2020YFSY0008), Young Scientist Program of Beijing Stomatological Hospital, and Capital Medical University (YSP202314).

Y.L. and Q.C. conceived the study. W.W. and J.L. performed laboratory assays and experiments, supervised by Y.L., W.W., J.L., and Y.D. analyzed the laboratory data. W.W., J.L., and B.T. produced the tables and figures. W.W. and J.L. wrote the first draft with assistance from Y.L. and Q.C. All authors critically reviewed and approved the final manuscript.

AUTHOR AFFILIATIONS

¹State Key Laboratory of Oral Diseases, National Clinical Research Center for Oral Diseases, Chinese Academy of Medical Sciences Research Unit of Oral Carcinogenesis and Management, West China Hospital of Stomatology, Sichuan University, Chengdu, Sichuan, China

²Department of Prosthodontics, Beijing Stomatological Hospital, School of Stomatology, Capital Medical University, Beijing, China

³CAS Key Laboratory of Environmental Biotechnology, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences, Beijing, China

AUTHOR ORCID*s*

Jing Li  <http://orcid.org/0009-0001-9663-8177>
Ye Deng  <http://orcid.org/0000-0002-7584-0632>
Yan Li  <http://orcid.org/0000-0002-2323-8105>
Qianming Chen  <http://orcid.org/0000-0001-8255-7635>

FUNDING

Funder	Grant(s)	Author(s)
MOST National Natural Science Foundation of China (NSFC)	81991500, 81991502	Qianming Chen
Key Projects of Sichuan Provincial Department of Science and Technology	2020YFSY0008	Yan Li
Young Scientist Program of Beijing Stomatological Hospital, Capital Medical University	YSP202314	Wei Wei
MOST National Natural Science Foundation of China (NSFC)	81771085	Yan Li

DATA AVAILABILITY

All data generated or analyzed during this study are included in this published article. The raw sequence data ([PRJCA017220](#)) of metagenomic and 16S rRNA reported in this paper have been deposited in the Genome Sequence Archive (Genomics, Proteomics & Bioinformatics 2021) in National Genomics Data Center (Nucleic Acids Res 2022), China National Center for Bioinformation / Beijing Institute of Genomics, Chinese Academy of Sciences (GSA: [CRA011605](#)) (GSA: [CRA011276](#)) that are publicly accessible at <https://ngdc.cncb.ac.cn/gsa>. The metabolomics data reported in this paper have been deposited in the OMIX, China National Center for Bioinformation / Beijing Institute of Genomics,

Chinese Academy of Sciences (<https://ngdc.cncb.ac.cn/omix>; accession no. OMIX004320) (46, 47).

ETHICS APPROVAL

This study was approved and supervised by the Medical Ethics Committee of West China Hospital of Stomatology, Sichuan University (WCHSIRB-OT 2019-015), and all participants gave informed consent. This animal study was approved by the Ethics Committee of West China Hospital of Stomatology, Sichuan University (WCHSIRB-D-2019-015).

ADDITIONAL FILES

The following material is available [online](#).

Supplemental Material

Supplemental material (mSystems00777-23-s0001.docx). Supplemental figures and table.

REFERENCES

- Chamoli A, Gosavi AS, Shirwadkar UP, Wangdale KV, Behera SK, Kurrey NK, Kalia K, Mandoli A. 2021. Overview of oral cavity squamous cell carcinoma: risk factors, mechanisms, and diagnostics. *Oral Oncol* 121:105451. <https://doi.org/10.1016/j.oraloncology.2021.105451>
- Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, Jemal A. 2018. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 68:394–424. <https://doi.org/10.3322/caac.21492>
- Chi AC, Day TA, Neville BW. 2015. Oral cavity and oropharyngeal squamous cell carcinoma—an update. *CA Cancer J Clin* 65:401–421. <https://doi.org/10.3322/caac.21293>
- Martínez-García M, Hernández-Lemus E. 2021. Periodontal inflammation and systemic diseases: an overview. *Front Physiol* 12:709438. <https://doi.org/10.3389/fphys.2021.709438>
- Shin YJ, Choung HW, Lee JH, Rhyu IC, Kim HD. 2019. Association of periodontitis with oral cancer: a case-control study. *J Dent Res* 98:526–533. <https://doi.org/10.1177/0022034519827565>
- Li T-J, Hao Y-H, Tang Y-L, Liang X-H. 2022. Periodontal pathogens: a crucial link between periodontal diseases and oral cancer. *Front Microbiol* 13:919633. <https://doi.org/10.3389/fmicb.2022.919633>
- Javed F, Warnakulasuriya S. 2016. Is there a relationship between periodontal disease and oral cancer? A systematic review of currently available evidence. *Crit Rev Oncol Hematol* 97:197–205. <https://doi.org/10.1016/j.critrevonc.2015.08.018>
- Komlós G, Csurgay K, Horváth F, Pelyhe L, Németh Z. 2021. Periodontitis as a risk for oral cancer: a case-control study. *BMC Oral Health* 21:640. <https://doi.org/10.1186/s12903-021-01998-y>
- Wei W, Li J, Shen X, Lyu J, Yan C, Tang B, Ma W, Xie H, Zhao L, Cheng L, Deng Y, Li Y, Gilbert JA, Thumbigere-Math V. 2022. Oral microbiota from periodontitis promote oral squamous cell carcinoma development via $\gamma\delta$ T cell activation. *mSystems* 7:e0046922. <https://doi.org/10.1128/mSystems.00469-22>
- Li L, Cao B, Liang X, Lu S, Luo H, Wang Z, Wang S, Jiang J, Lang J, Zhu G. 2019. Microenvironmental oxygen pressure orchestrates an anti- and pro-tumoral $\gamma\delta$ T cell equilibrium via tumor-derived exosomes. *Oncogene* 38:2830–2843. <https://doi.org/10.1038/s41388-018-0627-z>
- Rigau M, Ostrowska S, Fulford TS, Johnson DN, Woods K, Ruan Z, McWilliam HEG, Hudson C, Tutuka C, Wheatley AK, et al. 2020. Butyrophilin 2A1 is essential for phosphoantigen reactivity by $\gamma\delta$ T cells. *Science* 367:eaay5516. <https://doi.org/10.1126/science.aay5516>
- Jin C, Lagoudas GK, Zhao C, Bullman S, Bhutkar A, Hu B, Ameh S, Sandel D, Liang XS, Mazzilli S, Whary MT, Meyerson M, Germain R, Blainey PC, Fox JG, Jacks T. 2019. Commensal microbiota promote lung cancer development via $\gamma\delta$ T cells. *Cell* 176:998–1013. <https://doi.org/10.1016/j.cell.2018.12.040>
- Dupraz L, Magniez A, Rolhion N, Richard ML, Da Costa G, Touch S, Mayeur C, Planchais J, Agus A, Danne C, Michaudel C, Spatz M, Trottein F, Langella P, Sokol H, Michel M-L. 2021. Gut microbiota-derived short-chain fatty acids regulate IL-17 production by mouse and human intestinal $\gamma\delta$ T cells. *Cell Rep* 36:109332. <https://doi.org/10.1016/j.celrep.2021.109332>
- Helmink BA, Khan MAW, Hermann A, Gopalakrishnan V, Wargo JA. 2019. The microbiome, cancer, and cancer therapy. *Nat Med* 25:377–388. <https://doi.org/10.1038/s41591-019-0377-7>
- Liang D, Leung R-K, Guan W, Au WW. 2018. Involvement of gut microbiome in human health and disease: brief overview, knowledge gaps and research opportunities. *Gut Pathog* 10:3. <https://doi.org/10.1186/s13099-018-0230-4>
- Sami A, Elimairi I, Stanton C, Ross RP, Ryan CA. 2020. The role of the microbiome in oral squamous cell carcinoma with insight into the microbiome-treatment axis. *Int J Mol Sci* 21:8061. <https://doi.org/10.3390/ijms21218061>
- Zhao Z, Wu H, Jin T, Liu H, Men J, Cai G, Cernava T, Duan G, Jin D. 2023. Biodegradable mulch films significantly affected rhizosphere microbial communities and increased peanut yield. *Sci Total Environ* 871:162034. <https://doi.org/10.1016/j.scitotenv.2023.162034>
- Zhang C, Yin A, Li H, Wang R, Wu G, Shen J, Zhang M, Wang L, Hou Y, Ouyang H, et al. 2015. Dietary modulation of gut microbiota contributes to alleviation of both genetic and simple obesity in children. *EBioMedicine* 2:968–984. <https://doi.org/10.1016/j.ebiom.2015.07.007>
- Chen SY, Lin CH, Lin JT, Cheng YF, Chen HM, Kao SH. 2017. Adenine causes cell cycle arrest and autophagy of chronic myelogenous leukemia K562 cells via AMP-activated protein kinase signaling. *Oncol Lett* 14:5575–5580. <https://doi.org/10.3892/ol.2017.6890>
- Yamaguchi M, Murata T, Ramos JW. 2021. The phytochemical p-hydroxycinnamic acid suppresses the growth and stimulates the death in human liver cancer HepG2 cells. *Anticancer Drugs* 32:558–566. <https://doi.org/10.1097/CAD.0000000000001059>
- Bommagani S, Ponder J, Penthala NR, Janganani V, Jordan CT, Borrelli MJ, Crooks PA. 2017. Indole carboxylic acid esters of melampomagnolide B are potent anticancer agents against both hematological and solid tumor cells. *Eur J Med Chem* 136:393–405. <https://doi.org/10.1016/j.ejmech.2017.05.031>
- Hashemi M, Karami-Tehrani F, Ghavami S, Maddika S, Los M. 2005. Adenosine and deoxyadenosine induces apoptosis in oestrogen receptor-positive and -negative human breast cancer cells via the intrinsic pathway. *Cell Prolif* 38:269–285. <https://doi.org/10.1111/j.1365-2184.2005.00349.x>
- Lu'ng KVQ, Nguyễn LTH. 2013. The role of thiamine in cancer: possible genetic and cellular signaling mechanisms. *Cancer Genomics Proteomics* 10:169–185.

24. Sarasola M de la P, Táquez Delgado MA, Nicoud MB, Medina VA. 2021. Histamine in cancer immunology and immunotherapy. Current status and new perspectives. *Pharmacol Res Perspect* 9:e00778. <https://doi.org/10.1002/prp2.778>
25. Vitali R, Prioreschi C, Lorenzo Rebenaque L, Colantoni E, Giovannini D, Frusciante S, Diretto G, Marco-Jiménez F, Mancuso M, Casciati A, Pazzaglia S. 2022. Gut-brain axis: insights from hippocampal neurogenesis and brain tumor development in a mouse model of experimental colitis induced by dextran sodium sulfate. *Int J Mol Sci* 23:11495. <https://doi.org/10.3390/ijms231911495>
26. Zhao ZH, Xin FZ, Xue Y, Hu Z, Han Y, Ma F, Zhou D, Liu XL, Cui A, Liu Z, Liu Y, Gao J, Pan Q, Li Y, Fan JG. 2019. Indole-3-propionic acid inhibits gut dysbiosis and endotoxin leakage to attenuate steatohepatitis in rats. *Exp Mol Med* 51:1–14. <https://doi.org/10.1038/s12276-019-0304-5>
27. Cao Z, Ma J, Chen X, Zhou B, Cai C, Huang D, Zhang X, Cao D. 2016. Uridine homeostatic disorder leads to DNA damage and tumorigenesis. *Cancer Lett* 372:219–225. <https://doi.org/10.1016/j.canlet.2016.01.007>
28. Field MS, Lan X, Stover DM, Stover PJ. 2018. Dietary uridine decreases tumorigenesis in the *Ap^{Min/+}* model of intestinal cancer. *Curr Dev Nutr* 2:zy013. <https://doi.org/10.1093/cdn/nzy013>
29. Cao J, Balluff B, Arts M, Dubois LJ, van Loon LJC, Hackeng TM, van Eijk HHM, Eijkel G, Heij LR, Soons Z, Olde Damink SWM, Heeren RMA. 2021. Mass spectrometry imaging of L-[ring-13C6]-labeled phenylalanine and tyrosine kinetics in non-small cell lung carcinoma. *Cancer Metab* 9. <https://doi.org/10.1186/s40170-021-00262-9>
30. Bender MJ, McPherson AC, Phelps CM, Pandey SP, Laughlin CR, Shapira JH, Medina Sanchez L, Rana M, Richie TG, Mims TS, Gocher-Demske AM, Cervantes-Barragan L, Mullett SJ, Gelhaus SL, Bruno TC, Cannon N, McCulloch JA, Vignali DAA, Hinterleitner R, Joglekar AV, Pierre JF, Lee STM, Davar D, Zarour HM, Meisel M. 2023. Dietary tryptophan metabolite released by intratumoral *Lactobacillus reuteri* facilitates immune checkpoint inhibitor treatment. *Cell* 186:1846–1862. <https://doi.org/10.1016/j.cell.2023.03.011>
31. Messing SAJ, Ton-Hoang B, Hickman AB, McCubbin AJ, Peaslee GF, Ghirlando R, Chandler M, Dyda F. 2012. The processing of repetitive extragenic palindromes: the structure of a repetitive extragenic palindrome bound to its associated nuclease. *Nucleic Acids Res* 40:9964–9979. <https://doi.org/10.1093/nar/gks741>
32. Nunvar J, Huckova T, Licha I. 2010. Identification and characterization of repetitive extragenic palindromes (REP)-associated tyrosine transposases: implications for REP evolution and dynamics in bacterial genomes. *BMC Genomics* 11:44. <https://doi.org/10.1186/1471-2164-11-44>
33. Cosby RL, Judd J, Zhang R, Zhong A, Garry N, Pritham EJ, Feschotte C. 2021. Recurrent evolution of vertebrate transcription factors by transposase capture. *Science* 371:eabc6405. <https://doi.org/10.1126/science.abc6405>
34. Glenwright AJ, Pothula KR, Bhamidimarri SP, Chorev DS, Baslé A, Firbank SJ, Zheng H, Robinson CV, Winterhalter M, Kleinekathöfer U, Bolam DN, van den Berg B. 2017. Structural basis for nutrient acquisition by dominant members of the human gut microbiota. *Nature* 541:407–411. <https://doi.org/10.1038/nature20828>
35. Reeves AR, D'Elia JN, Frias J, Salyers AA. 1996. A *Bacteroides thetaiotaomicron* outer membrane protein that is essential for utilization of maltooligosaccharides and starch. *J Bacteriol* 178:823–830. <https://doi.org/10.1128/jb.178.3.823-830.1996>
36. Huang J-Y, Lin Y-C, Chen H-M, Lin J-T, Kao S-H. 2022. Adenine combined with cisplatin promotes anticancer activity against hepatocellular cancer cells through AMPK-mediated p53/p21 and p38 MAPK cascades. *Pharmaceuticals (Basel)* 15:795. <https://doi.org/10.3390/ph15070795>
37. Hoffman JA, Badger JL, Zhang Y, Huang SH, Kim KS. 2000. *Escherichia coli* K1 aslA contributes to invasion of brain microvascular endothelial cells *in vitro* and *in vivo*. *Infect Immun* 68:5062–5067. <https://doi.org/10.1128/IAI.68.9.5062-5067.2000>
38. Kim KS. 2002. Strategy of *Escherichia coli* for crossing the blood-brain barrier. *J Infect Dis* 186 Suppl 2:S220–4. <https://doi.org/10.1086/344284>
39. Patil RS, Shah SU, Shrikhande SV, Goel M, Dikshit RP, Chiplunkar SV. 2016. IL17 producing $\gamma\delta$ T cells induce angiogenesis and are associated with poor survival in gallbladder cancer patients. *Int J Cancer* 139:869–881. <https://doi.org/10.1002/ijc.30134>
40. Peng G, Wang HY, Peng W, Kiniwa Y, Seo KH, Wang RF. 2007. Tumor-infiltrating gammadelta T cells suppress T and dendritic cell function via mechanisms controlled by a unique toll-like receptor signaling pathway. *Immunity* 27:334–348. <https://doi.org/10.1016/j.immuni.2007.05.020>
41. Daley D, Zambirinis CP, Seifert L, Akkad N, Mohan N, Werba G, Barilla R, Torres-Hernandez A, Hundeyin M, Kumar Mani VR, Avanzi A, Tippens D, Narayanan R, Jang J-E, Newman E, Pillarisetty VG, Dustin ML, Bar-Sagi D, Hajdu C, Miller G. 2020. $\gamma\delta$ T cells support pancreatic oncogenesis by restraining $\alpha\beta$ T cell activation. *Cell* 183:1134–1136. <https://doi.org/10.1016/j.cell.2020.10.041>
42. Charnavets T, Nunvar J, Nečasová I, Völker J, Breslauer KJ, Schneider B. 2015. Conformational diversity of single-stranded DNA from bacterial repetitive extragenic palindromes: implications for the DNA recognition elements of transposases. *Biopolymers* 103:585–596. <https://doi.org/10.1002/bip.22666>
43. Gilson E, Saurin W, Perrin D, Bachellier S, Hofnung M. 1991. The BIME family of bacterial highly repetitive sequences. *Res Microbiol* 142:217–222. [https://doi.org/10.1016/0923-2508\(91\)90033-7](https://doi.org/10.1016/0923-2508(91)90033-7)
44. Shi R, Li Y, Liu H, Chen J. 2021. Lung microbiota: unexploited treasure hidden in the immune microenvironment of lung cancer. *Thorac Cancer* 12:2964–2966. <https://doi.org/10.1111/1759-7714.14159>
45. Fleming C, Cai Y, Sun X, Jala VR, Xue F, Morrissey S, Wei YL, Chien YH, Zhang HG, Haribabu B, Huang J, Yan J. 2017. Microbiota-activated CD103⁺ DCs stemming from microbiota adaptation specifically drive $\gamma\delta$ T17 proliferation and activation. *Microbiome* 5:46. <https://doi.org/10.1186/s40168-017-0263-9>
46. CNCR-NGDC Members and Partners. 2022. Database resources of the national genomics data center, China national center for bioinformatics in 2022. *Nucleic Acids Res* 50:D27–D38. <https://doi.org/10.1093/nar/gkab951>
47. Chen T, Chen X, Zhang S, Zhu J, Tang B, Wang A, Dong L, Zhang Z, Yu C, Sun Y, Chi L, Chen H, Zhai S, Sun Y, Lan L, Zhang X, Xiao J, Bao Y, Wang Y, Zhang Z, Zhao W. 2021. The genome sequence archive family: toward explosive data growth and diverse data types. *Genomics Proteomics Bioinformatics* 19:578–583. <https://doi.org/10.1016/j.gpb.2021.08.001>